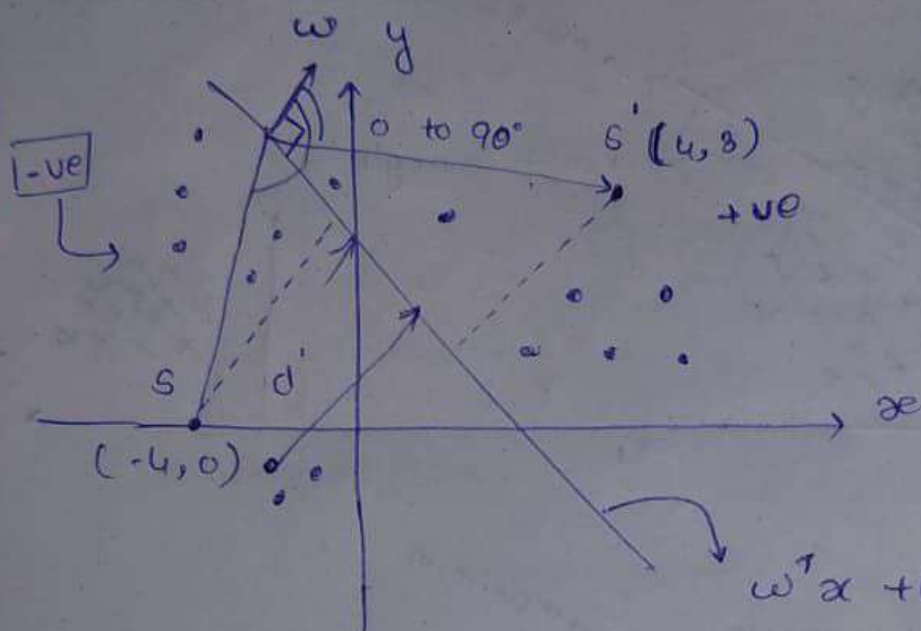


⊛ Support Vector Machines (SVC)



Math Intuition

$$ax + by + c = 0$$

$$w_1 x_1 + w_2 x_2 + b = 0$$

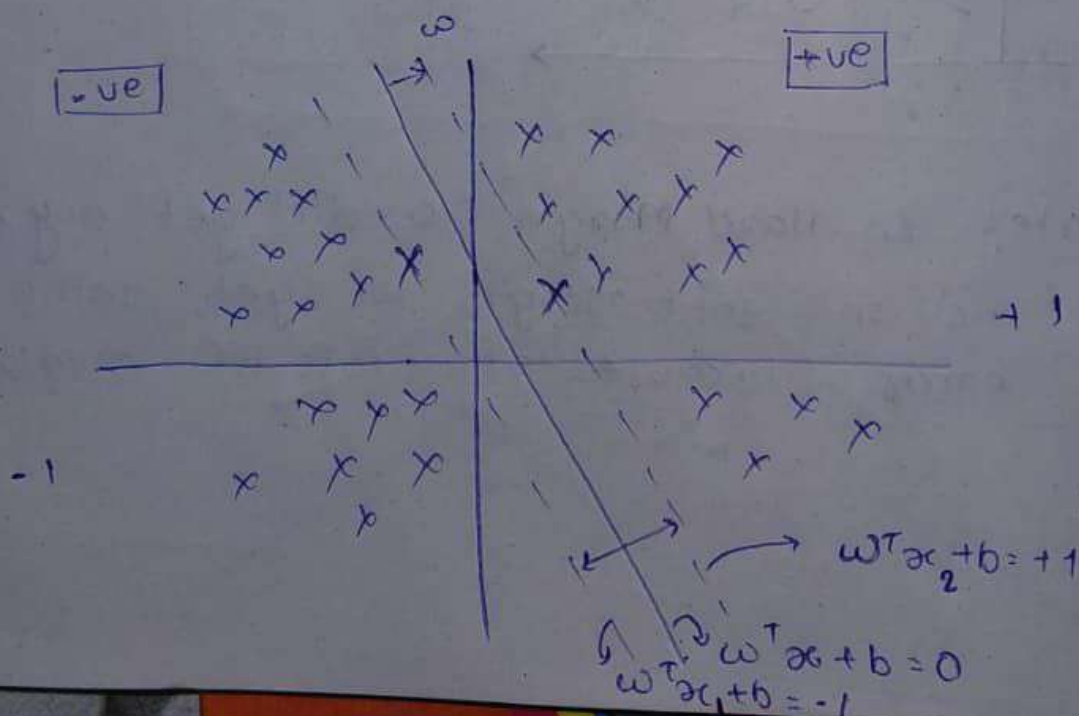
$$w^T x + b = 0$$

$$b = 0$$

$$w^T x = 0$$

$d = -ve$ below plane

$d = +ve$ above plane



$$\begin{aligned} w^T x_1 + b &= 1 \\ (-) w^T x_2 + b &= (-1) \end{aligned}$$

$$\frac{w^T (x_1 - x_2)}{\|w\|} = \frac{+2}{\|w\|}$$

unit vector & magnitude of the vector is 1

cost function

Maximize $\frac{2}{\|w\|} \Rightarrow$ Distance between
 w, b marginal plane

constant such that $\Rightarrow$ correctly classified points

$$y_i \left\{ \begin{array}{ll} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{array} \right\}$$

for all correct points

constraint $\rightarrow$ $y_i * (w^T x + b) \geq 1$